# Dual-Subspace Motion Modeling for Robust Dynamic Gaussian Splatting

Bingbing Hu[1†], Yan Li[2†], Yong Yang[1], Shihui Guo[1], Junfeng Yao[1,3*], Hesheng Wang[4*]

[1*]School of Film, School of Informatics, Xiamen University, China.
[2]School of Computing, National University of Singapore, Singapore.
[3]Institute of Artificial Intelligence, Xiamen University, China.
[4]Department of Automation, Shanghai Jiao Tong University, China.

*Corresponding author(s). E-mail(s): yao0010@xmu.edu.cn; wanghesheng@sjtu.edu.cn;
Contributing authors: hubingbing@stu.xmu.edu.cn; yan.li@nus.edu.sg;
yangyongzp@stu.xmu.edu.cn; guoshihui@xmu.edu.cn;
†These authors contributed equally to this work.

**Abstract**

Reconstructing dynamic scenes from monocular videos remains a challenging problem due to motion modeling methods generally suffering from temporal instability, boundary artifacts, or geometric collapse. To address this issue, we propose a principled motion decomposition framework for monocular dynamic scene reconstruction based on dynamic Gaussian Splatting. Our method explicitly separates motion into two complementary components: a Chebyshev field that models globally stable, low-frequency motion trends, and a Residual Motion field that captures localized, high-frequency dynamics outside the spectral subspace. The novel Chebyshev field module leverages spectral representations on bounded, non-periodic temporal intervals to ensure temporal coherence and suppress boundary oscillations, while the residual motion field focuses on structured local dynamics without interfering with global stability. We further design a joint optimization objective that enforces appearance consistency, geometric plausibility, and residual regularization, ensuring stable and physically meaningful reconstruction. Extensive experiments on challenging monocular dynamic scenes demonstrate that the proposed framework significantly improves temporal stability, reconstruction fidelity, and motion boundary sharpness compared to existing dynamic Gaussian and neural deformation methods.

**Keywords:** Dynamic Gaussian Splatting, Chebyshev Polynomials

## 1 Introduction

Reconstructing high-fidelity dynamic scenes from monocular videos [52, 10, 28] remains a fundamental challenge in open-world settings. Without multi-view [12, 26] or depth supervision [55, 27, 50, 49], a model must simultaneously infer temporally evolving geometry and appearance from highly ambiguous observations. Although recent advances in Neural Radiance Fields (NeRFs) [8, 3, 37] and Gaussian-based representations [4, 5, 44, 25, 23, 24] have significantly improved rendering quality, robust motion modeling in monocular

dynamic scenes remains elusive. In practice, many methods still suffer from temporal jitter, boundary instability, or geometry collapse when confronted with complex non-rigid motion [21, 23, 10].

A central difficulty lies in an inherent *stability–plasticity dilemma* [8, 3, 33, 35]. On the one hand, motion must remain physically coherent and temporally stable to support consistent reconstruction. On the other hand, real-world scenes exhibit highly non-linear, localized, and semantic-dependent motion patterns that demand substantial modeling flexibility. Balancing these two requirements is particularly challenging in monocular settings, where photometric supervision alone provides limited geometric constraints.

Current approaches [52, 44, 13] to this problem typically grapple with a fundamental theoretical trade-off. Existing methods generally fall into two categories, each prioritizing one aspect at the expense of the other. **Implicit Deformation Fields**, such as D3DGS and 4DGS [52, 44], rely on the universal approximation capabilities of Multi-Layer Perceptrons (MLPs) to model motion. While offering high *plasticity* to fit arbitrary deformations, these methods are governed by the Neural Tangent Kernel (NTK) theory, which dictates that coordinate-based networks exhibit a strong **spectral bias** [36]. Consequently, they tend to over-smooth high-frequency transient details, such as rapid jitter or sudden texture changes, resulting in blurry reconstructions. Conversely, **Explicit Trajectory Models**, employ polynomial or Fourier bases [16, 26] to parameterize motion. These bases provide strong temporal smoothness priors, ensuring *stability*. However, the linear space spanned by these explicit functions is fundamentally limited. Complex non-linear dynamics, such as topological breaks (e.g., object separation) or intricate semantic interactions (e.g., cloth wrinkling), often reside in the **null space** of the basis functions. Failing to capture these dynamics leads to underfitting or the generation of inexplicable `floater` artifacts.

In this work, we challenge this assumption and argue that dynamic motion in real-world scenes does not reside in a single function space. Instead, it naturally decomposes into components with fundamentally different structural properties. Global motion trends are typically low-frequency, temporally coherent, and physically constrained, whereas localized motion is often high-frequency, irregular, and spatially heterogeneous. Treating these components uniformly inevitably sacrifices either stability or flexibility.

Motivated by this observation, we propose a principled motion decomposition framework for monocular dynamic scene reconstruction based on dynamic Gaussian splatting. Our method explicitly separates motion into two complementary components: a *Chebyshev field* that models globally stable, low-frequency motion trends, and a *Residual Motion Field* that captures localized, high-frequency dynamics outside the spectral subspace. This decomposition introduces inductive bias for temporal coherence while preserving the flexibility required to model complex non-rigid motion. The Chebyshev field leverages spectral representations on bounded, non-periodic intervals to provide stable global motion modeling and suppress boundary artifacts commonly observed in polynomial [12] or Fourier-based trajectories [16]. By contrast, the residual motion field operates exclusively in the spectral complement, focusing its capacity on structured local dynamics without interfering with global stability. Together, these components resolve the stability–plasticity dilemma inherent in monolithic deformation models.

The main contributions of this work are summarized as follows:

- We identify the stability–plasticity dilemma in monocular dynamic reconstruction as a consequence of monolithic motion modeling and propose a principled decomposition-based perspective.
- We introduce a dynamic Gaussian splatting framework that explicitly separates motion into a globally stable Chebyshev field and a localized residual motion field.
- We design a joint optimization strategy that respects the complementary roles of the two motion components, leading to stable and physically plausible reconstruction.
- We demonstrate state-of-the-art performance on challenging dynamic scenes, achieving improved stability and reconstruction quality over existing methods.

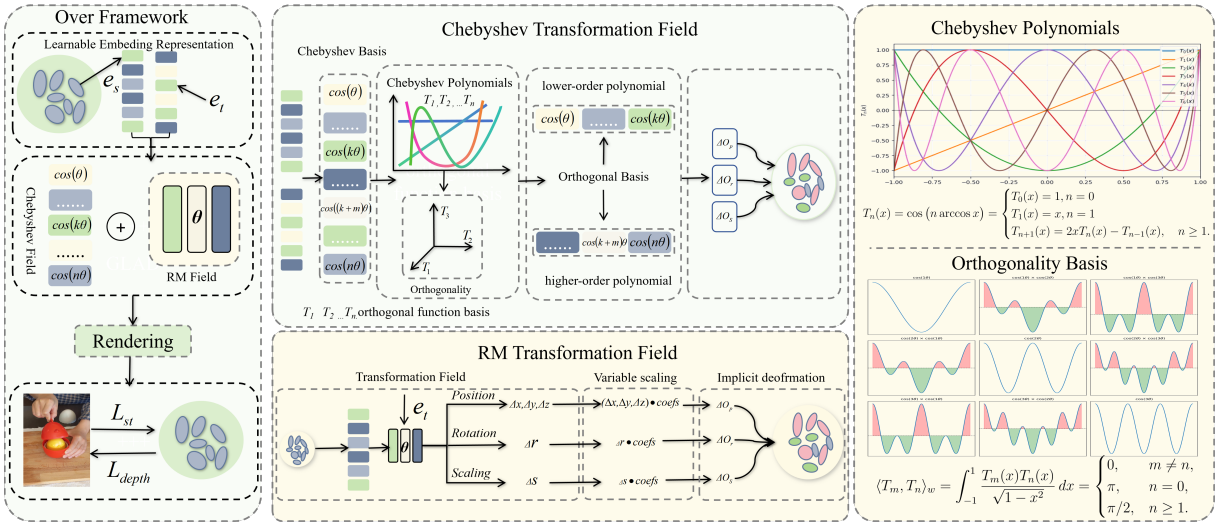



**Fig. 1**: **Overview of the proposed framework**. Our method adopts a divide-and-conquer strategy for monocular dynamic scene reconstruction by explicitly decomposing motion into two complementary transformation fields. The Chebyshev Transformation Field models globally coherent and temporally stable motion using a low-frequency spectral representation based on Chebyshev polynomial bases, providing a robust backbone for bounded, non-periodic motion. In parallel, the Residual Motion (RM) Field, implemented as a lightweight implicit model, captures localized and high-frequency motion components that lie outside the Chebyshev spectral subspace.

# 2 Related Work

## 2.1 NeRF for Dynamic Scenes

Neural Radiance Fields for Dynamic Scenes. The advent of Neural Radiance Fields (NeRF) [32, 2, 8, 3, 37] has fundamentally shifted the paradigm of view synthesis, enabling photorealistic reconstruction of static scenes via continuous implicit representations. However, extending this capability to dynamic scenes presents a non-trivial challenge, particularly when supervision is limited to sparse camera views. To bridge the gap between static and time-varying reconstruction, recent literature has predominantly diverged into two methodological streams. The first category of approaches introduces deformation fields to model motion. These methods [34, 11] typically learn a mapping function that warps spatiotemporal coordinates from a dynamic observation space into a canonical static space. By disentangling geometry from motion, they enforce a structural bias that helps maintain consistency. However, estimating complex non-rigid deformations from sparse inputs remains an ill-posed problem, often leading to artifacts in regions with rapid motion. The second category focuses on representation efficiency by decomposing the 4D volume. Recognizing the computational bottleneck of pure MLP-based networks, recent works such as K-Planes [8], HexPlane [3], and Tensor4D [37] propose to project the 4D spatiotemporal field into lower-dimensional subspaces (e.g., planar or tensor grids). These explicit or hybrid representations significantly accelerate training and inference. However, in sparse view scenarios, the lack of geometric constraints causes these discretized representations to produce "floater" artifacts or blurriness, as the model tends to overfit to the few available training frames rather than learning the underlying scene geometry.

## 2.2 3D Gaussian Splatting for Dynamic Scenes

***Dynamic Gaussian Splatting.***

Unlike NeRFs which implicitly encode scenes within neural weights, 3D Gaussian Splatting (3DGS) [17, 24, 38, 53] utilizes explicit point-based primitives to enable real-time photorealistic rendering. The transition to dynamic 3DGS [51, 10] has spurred a variety of strategies to model

time-dependent geometry. Initial attempts, such as D3DGS [52] and 4DGS [44], borrowed concepts from dynamic NeRFs, utilizing deformation networks or hex-plane factorizations to modulate Gaussian attributes. However, to better align with the explicit nature of Gaussians, subsequent research has explored trajectory-based and physically-grounded formulations. STGS [26] models Gaussian motion through polynomial trajectories, while SC-GS [14] enforces rigidity via Linear Blend Skinning (LBS) and control points. Pushing this explicitness further, Ex4DGS [20] adopts a strategy of separating static and dynamic elements, employing sparse timestamp sampling and interpolation to reconstruct motion tracks. However, these distinct modeling paradigms all face inherent limitations in sparse view scenarios. MLP-based deformation networks (e.g., D3DGS [52] and 4DGS [44]) effectively act as black-box approximators; without dense supervision, they tend to learn arbitrary, physically implausible warpings to satisfy training pixels, leading to high-frequency artifacts in novel views. Conversely, methods relying on polynomial trajectories or explicit interpolation are often bounded by their fixed basis functions or linear assumptions. Under sparse constraints, these rigid representations struggle to capture complex non-linear dynamics, often resulting in either over-smoothed motion or spurious oscillations. Consequently, both streams of methods are prone to geometric collapse and severe `floater` artifacts when observing dynamic scenes from limited angles.

***Recent dynamic Gaussian Splatting methods.***

Recent dynamic Gaussian Splatting methods have further improved the quality, efficiency, and scalability of dynamic scene reconstruction. Representative methods such as MoDec-GS [19], FRoG [15], 4K4D [46], LongVol-TemporalGH [47], Swift4D [45], DeGauss [42], and FreeTimeGS [43] explore different aspects of dynamic Gaussian representations, including efficient rendering, long-video modeling, temporal regularization, dynamic/static decomposition, and scalable dynamic view synthesis. These methods significantly advance dynamic Gaussian Splatting from different perspectives and provide strong baselines for evaluating recent progress in this area.

Different from these methods, our work focuses on temporal motion decomposition for monocular dynamic reconstruction. Instead of only separating static and dynamic regions or reducing redundant Gaussian primitives, we decompose the deformation field into a globally stable low-order Chebyshev component and a complementary RM residual component. The Chebyshev branch captures smooth temporal motion trends, while the RM branch models localized residual deformation that is difficult to represent with low-order temporal bases alone. Therefore, our method is complementary to recent efficient dynamic Gaussian Splatting methods, as it improves the structure of the motion representation rather than only optimizing representation size or rendering efficiency.

# 3 Preliminary: 3D Gaussian Splatting

Instead of dense volumetric sampling, we employ a collection of explicit 3D Gaussian [17, 41] primitives to encapsulate both the structural and photometric attributes of the scene. This strategy enables compact storage and facilitates high-speed differentiable rasterization. Mathematically, the shape of the $i$-th element is defined by:

$$G_i(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right). \quad (1)$$

The primitive is centered at $\boldsymbol{\mu}_i$. Its anisotropic shape is determined by the covariance matrix $\boldsymbol{\Sigma}$, which is decomposed into a rotation matrix $\mathbf{R}$ and a diagonal scaling matrix $\mathbf{S}$ (i.e., $\boldsymbol{\Sigma} = \mathbf{RSS}^\top\mathbf{R}^\top$) to guarantee physical validity during optimization. For rasterization, the 3D primitives are projected onto the screen space and integrated to form the final image. The resulting pixel color $C$ is derived through an ordered volumetric compositing equation:

$$C = \sum_{i\in\mathcal{N}} c_i, \alpha_i \prod_{j=1}^{i-1}(1-\alpha_j), \quad (2)$$

where $c_i$ is the view-dependent radiance calculated from spherical harmonics coefficients, and $\alpha_i$ indicates the contributing opacity. Crucially, the blending process strictly follows the depth order of

Gaussians relative to the camera to ensure correct visibility.

# 4 Methodology

## 4.1 Problem Formulation: Stability–Plasticity Dilemma

Recovering high-fidelity dynamic scenes from monocular videos is fundamentally ill-posed, as it requires modeling temporally evolving geometry and appearance without multi-view constraints. A central challenge lies in the inherent *stability-plasticity dilemma*: the motion field must remain physically coherent and temporally stable, while simultaneously retaining sufficient flexibility to represent highly nonlinear local deformations.

Existing approaches, as illustrated in Fig. 2, predominantly model temporal dynamics as a *monolithic function*, which inevitably leads to incomplete trade-offs between stability and expressiveness. Implicit deformation fields parameterized by MLPs offer strong representational capacity; however, as shown in Fig. 2a, their frame-wise optimization often results in temporal jitter and unstable trajectories due to gradient conflicts across time. Conversely, explicit trajectory models based on predefined temporal bases enforce strong smoothness priors and yield stable global trends, but their limited function span causes systematic underfitting of local variations, as exemplified by the Chebyshev-only case in Fig. 2d. Other explicit designs suffer from distinct but related failure modes: polynomial or Fourier bases introduce severe boundary oscillations (Fig. 2b), while temporal RBF models fragment motion into locally smooth yet globally inconsistent segments (Fig. 2c).

These observations suggest that the above limitations are not artifacts of specific architectures or optimization strategies, but rather stem from a flawed modeling assumption: *dynamic motion in real-world scenes does not reside in a single function space*. Instead, motion naturally decomposes into complementary components with fundamentally different structural properties. **Global motion** trends are typically low-frequency, temporally coherent, and physically constrained, whereas **local deformations** are generally high-frequency, spatially localized, and involve complex non-rigid dynamics.

Motivated by this insight, we reformulate dynamic scene modeling as a divide-and-conquer problem in function space, as conceptually illustrated in Fig. 1. Specifically, the time-varying state of a Gaussian primitive $G(t)$ is decomposed into two complementary subspaces:

$$G(t) = G(0) + \Phi_{\text{Cheb}}(t) + \Psi_{\text{RM}}(t), \tag{3}$$

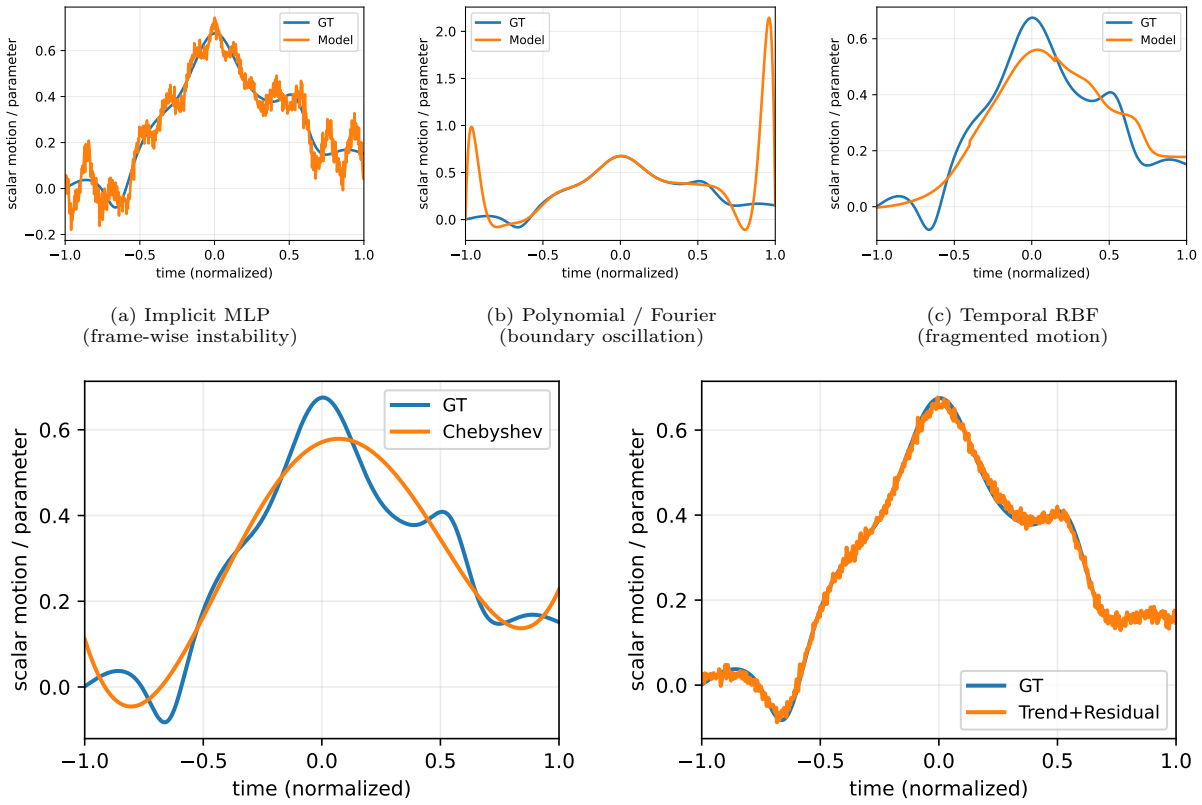
where $\Phi_{\text{Cheb}}$ captures spectrally structured, globally smooth motion trends, and $\Psi_{\text{RM}}$ models residual motion deformations that cannot be explained by such global dynamics. As illustrated in Fig. 2(b), the combination of these two components enables simultaneous global stability and local fidelity. This decomposition introduces a strong inductive bias that explicitly separates stability-critical dynamics from highly nonlinear variations, enabling robust optimization while preserving expressive power.

## 4.2 The Chebyshev Field: Spectrally Stable Global Motion Modeling

Following the motion decomposition introduced in Eq. 3, the Chebyshev field $\Phi_{\text{Cheb}}$ is designed to model the *globally coherent and temporally stable* component of dynamic motion. As illustrated in Fig. 2d, this component corresponds to low-frequency motion trends that are physically meaningful and consistent over time, such as rigid or near-rigid motion and smooth articulation.

A key requirement for modeling such global dynamics is *temporal stability on bounded, non-periodic intervals*. In practical dynamic scene reconstruction, motion is observed within a finite temporal window and rarely exhibits periodic behavior. Spectral representations that implicitly assume periodicity, such as Fourier bases, therefore tend to introduce boundary oscillations and ringing artifacts near temporal endpoints (Fig. 2b), which directly translate into geometric instability.

To address this issue, we adopt Chebyshev polynomials of the first kind as the spectral basis for global motion modeling. Chebyshev polynomials are defined on bounded intervals and provide near-minimax approximation properties under non-uniform sampling, yielding uniformly controlled approximation error across time. These




(a) Implicit MLP (frame-wise instability)

(b) Polynomial / Fourier (boundary oscillation)

(c) Temporal RBF (fragmented motion)

(d) Ours (Chebyshev-only)

(e) Ours (Chebyshev trend + residual)

**Fig. 2**: Comparison of temporal modeling strategies on a representative 1D motion signal. (a) Implicit MLP-based modeling exhibits frame-wise instability and temporal jitter due to optimization conflicts. (b) Polynomial or Fourier-based representations suffer from boundary oscillations near temporal endpoints. (c) Temporal RBF modeling leads to fragmented motion with limited global coherence. (d) Chebyshev-only spectral modeling provides stable and smooth global motion trends but underfits localized variations. (e) Our full model combines a Chebyshev-based global trend with a residual motion component, achieving both temporal stability and accurate reconstruction of local dynamics.

properties make Chebyshev representations particularly suitable for stable modeling of non-periodic motion trends.

#### 4.2.1 Chebyshev-Based Spectral Parameterization

Given a normalized temporal coordinate $\tau(t) \in [-1, 1]$, we approximate smooth global motion using a truncated Chebyshev expansion:

$$f(t) \approx \sum_{k=0}^{D} c_k \, T_k(\tau(t)), \tag{4}$$

$$\begin{aligned} T_n(x) &= \cos\big(n \arccos x\big) \\ &= \begin{cases} 1, n = 0 \\ x, n = 1 \\ 2xT_{n-1}(x) - T_{n-2}(x), \quad n \geq 2. \end{cases} \end{aligned} \tag{5}$$

where $T_k(\cdot)$ denotes the $k$-th Chebyshev polynomial of the first kind and $\{c_k\}$ are learnable spectral coefficients.

This representation provides two properties critical for global motion modeling. First, Chebyshev polynomials form an orthogonal basis under

a weighted inner product, enabling a clear separation between low- and high-frequency components. Second, their non-uniform node distribution concentrates approximation capacity near temporal boundaries, significantly suppressing oscillations compared to uniform polynomial interpolation. As a result, Chebyshev-based representations yield smooth and stable motion trajectories over bounded temporal windows.

**Orthogonality of Chebyshev Bases**. Chebyshev polynomials satisfy orthogonality in a weighted inner product space:

$$\begin{aligned}\langle T_m, T_n \rangle_w &= \int_{-1}^{1} \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}}\, dx \\ &= \begin{cases} 0, & m \neq n, \\ \pi, & m = n = 0, \\ \pi/2, & m = n \geq 1 \end{cases} \end{aligned} \tag{6}$$

This orthogonality property forms the mathematical foundation of our "decoupling" mechanism. It implies that motion components at different frequencies are statistically independent.

**Theoretical Justification: Error Bound Analysis**

Our choice of the Chebyshev field is theoretically grounded in the minimax property of polynomial interpolation. Consider the temporal domain $t \in [-1, 1]$. When approximating a smooth motion trajectory $f(t)$ using a polynomial $P_{N-1}(t)$ of degree $N-1$ based on $N$ interpolation nodes $\{t_i\}_{i=1}^{N}$, the interpolation error is given by the Lagrange remainder theorem:

$$E(t) = f(t) - P_{N-1}(t) = \frac{f^{(N)}(\xi)}{N!} \prod_{i=1}^{N} (t - t_i), \tag{7}$$

where $\xi \in [-1, 1]$ and $f^{(N)}$ denotes the $N$-th derivative. The term $\omega(t) = \prod_{i=1}^{N}(t - t_i)$ is a monic polynomial of degree $N$, solely determined by the node distribution. To minimize the global approximation error, specifically the worst-case error at the boundaries, we seek to minimize the infinity norm $\|\omega(t)\|_\infty$.

Chebyshev polynomials of the first kind, defined as $T_N(t) = \cos(N \arccos t)$, have a leading coefficient of $2^{N-1}$. Consequently, the normalized polynomial $\tilde{T}_N(t) = 2^{-(N-1)} T_N(t)$ is a monic polynomial with the unique property of having the minimal infinity norm on $[-1, 1]$. Since $|T_N(t)| \leq 1$, the bound for the nodal polynomial becomes:

$$\min_{\{t_i\}} \|\omega(t)\|_\infty = \|\tilde{T}_N(t)\|_\infty = \frac{1}{2^{N-1}}. \tag{8}$$

This derivation proves that by aligning the field representation with Chebyshev nodes, the interpolation error is bounded by a factor that decays exponentially with $N$. This stands in stark contrast to equidistant sampling, where the Lebesgue constant grows exponentially, causing the Runge phenomenon. Thus, our Chebyshev field theoretically guarantees stability for modeling non-periodic global motion trends.

#### 4.2.2 Chebyshev as a Continuous Transformation Field

Building upon the spectral parameterization above, we construct the Chebyshev field as a continuous transformation field that directly governs the temporal evolution of Gaussian primitives:

$$\Phi_{\text{Cheb}} : (e_s, e_t) \rightarrow \Delta G(t), \tag{9}$$

where $e_s$ denotes the spatial embedding associated with each Gaussian primitive and $e_t$ represents its temporal embedding. The predicted deformation $\Delta G(t)$ modulates the translation, rotation, and scale parameters of Gaussian primitives.

In practice, the Chebyshev field is optimized jointly with Gaussian parameters in an end-to-end differentiable Gaussian splatting pipeline. By explicitly modeling globally smooth motion trends, $\Phi_{\text{Cheb}}$ provides a stable deformation backbone shared across time, which serves as a foundation for subsequent residual refinement.

### 4.3 Residual Motion Field: Local and High-Frequency Dynamics

While the Chebyshev field explicitly models globally coherent and low-frequency motion trends, real-world dynamic scenes inevitably contain motion components that cannot be captured by a global spectral representation alone. These components include localized non-rigid deformation, transient object interactions, and irregular motion patterns that vary across space and time.

Crucially, such dynamics are not random noise but structured residual motion that lies outside

the Chebyshev spectral subspace. Forcing these components into the global spectral model would either require excessively high polynomial degrees, reintroducing unwanted global oscillations, or lead to systematic underfitting. We therefore introduce a complementary *Residual Motion Field* (RM field) to explicitly model these high-frequency and localized motion components.

#### 4.3.1 Residual Motion in the Spectral Complement

Let $M_{\text{true}}(t)$ denote the underlying motion of a Gaussian primitive and $P_{\text{Cheb}}(\cdot)$ the projection induced by the Chebyshev field. The residual motion component is defined as:

$$R(t) = M_{\text{true}}(t) - P_{\text{Cheb}}\big(M_{\text{true}}(t)\big). \tag{10}$$

By construction, $R(t)$ lies in the orthogonal complement of the Chebyshev spectral subspace and predominantly consists of high-frequency, localized motion patterns. The RM field $\Psi_{\text{RM}}$ is designed to approximate this residual component:

$$\Psi_{\text{RM}}(e_s, e_t) \approx R(t). \tag{11}$$

Unlike monolithic implicit deformation models that attempt to approximate the entire motion field with a single function, the RM field operates exclusively on the residual space left unexplained by the global spectral model. This separation allows the model to allocate representational capacity where it is most needed, without interfering with globally stable motion trends.

#### 4.3.2 Spatio-Temporal Conditioning for Residual Modeling

Effective residual modeling requires conditioning on both spatial identity and temporal state. We therefore parameterize the RM field using a dual conditioning mechanism composed of spatial and temporal embeddings.

***Spatial Embedding.***

Each Gaussian primitive is associated with a learnable spatial embedding $e_s \in \mathbb{R}^{32}$, which encodes its local geometric and motion identity. Primitives undergoing similar residual dynamics tend to develop correlated embeddings, enabling coherent deformation for articulated or semi-rigid structures while preserving separability across distinct objects.

***Continuous Temporal Embedding.***

To support continuous-time motion modeling and irregular temporal sampling, we adopt a learnable temporal feature grid $\gamma$, from which the temporal embedding $e_t$ is queried via interpolation:

$$e_t = \text{Interp}(\gamma, t). \tag{12}$$

This continuous representation provides smooth temporal conditioning and avoids discretization artifacts associated with frame-indexed embeddings.

#### 4.3.3 Residual Prediction and Frequency Handoff

The RM field is parameterized as a lightweight multi-layer perceptron $g_\theta$, which takes the concatenated conditioning vector $[e_s; e_t]$ as input and predicts residual offsets for translation, rotation, and scale:

$$\Delta a_i = g_\theta\big([e_s; e_t]_i\big). \tag{13}$$

A key advantage of the proposed design lies in the explicit *frequency handoff* between the Chebyshev and RM fields. Since low-frequency global motion trends are already captured by the Chebyshev field, the RM field is relieved from approximating these components and can dedicate its full capacity to high-frequency residual dynamics. This separation mitigates the spectral bias [36] commonly observed in implicit deformation models and leads to more stable optimization and sharper motion boundaries, as illustrated in Fig. 2e.

### 4.4 Diagnostic formulation for branch complementarity.

To verify whether the RM branch learns complementary residual motion rather than duplicating the Chebyshev branch, we use a projection-based diagnostic. Following the dynamic state decomposition in Eq. (3), we analyze the temporal deformation signals of the Chebyshev branch $\Phi_{\text{Cheb}}$, the RM branch $\Psi_{\text{RM}}$, and the full deformation $\Delta G = \Phi_{\text{Cheb}} + \Psi_{\text{RM}}$.

Given the normalized time coordinate $\tau(t) \in [-1, 1]$, we construct the degree-$D$ Chebyshev basis matrix

$$\mathbf{B}_{\text{Cheb}} = [T_0(\tau), T_1(\tau), \ldots, T_D(\tau)] \in \mathbb{R}^{T\times(D+1)}, \tag{14}$$

and define the projection matrix onto the Chebyshev temporal subspace as

$$\mathbf{P}_{\text{Cheb}} = \mathbf{B}_{\text{Cheb}}\mathbf{B}^{+}_{\text{Cheb}}, \tag{15}$$

where $\mathbf{B}^{+}_{\text{Cheb}}$ denotes the Moore–Penrose pseudo-inverse of $\mathbf{B}_{\text{Cheb}}$. When $\mathbf{B}^{\top}_{\text{Cheb}}\mathbf{B}_{\text{Cheb}}$ is full-rank, it reduces to

$$\mathbf{B}^{+}_{\text{Cheb}} = \left(\mathbf{B}^{\top}_{\text{Cheb}}\mathbf{B}_{\text{Cheb}}\right)^{-1}\mathbf{B}^{\top}_{\text{Cheb}}. \tag{16}$$

For a sampled temporal deformation signal $\mathbf{S} \in \mathbb{R}^{T\times 3N_g}$, we decompose it into the Chebyshev-subspace component and the complementary component:

$$\mathbf{S}^{\text{Cheb}} = \mathbf{P}_{\text{Cheb}}\mathbf{S}, \quad \mathbf{S}^{\perp} = (\mathbf{I} - \mathbf{P}_{\text{Cheb}})\mathbf{S}. \tag{17}$$

The corresponding energy ratios are defined as

$$\rho_{\text{Cheb}}(\mathbf{S}) = \frac{\|\mathbf{S}^{\text{Cheb}}\|_F^2}{\|\mathbf{S}\|_F^2 + \epsilon}, \quad \rho_{\perp}(\mathbf{S}) = \frac{\|\mathbf{S}^{\perp}\|_F^2}{\|\mathbf{S}\|_F^2 + \epsilon}. \tag{18}$$

For the RM branch, $\rho_{\text{Cheb}}(\Psi_{\text{RM}})$ measures the projection leakage into the Chebyshev temporal subspace, while $\rho_{\perp}(\Psi_{\text{RM}})$ measures the residual energy outside this subspace. These diagnostics provide empirical evidence for the complementary behavior of the two branches.

## 4.5 Joint Optimization Objective

Since recovering 4D dynamics from monocular video is fundamentally an ill-posed problem, relying solely on RGB supervision often leads to degenerate solutions where geometry collapses to explain appearance changes. To regularize the optimization landscape, we design a composite objective function that imposes constraints across three distinct domains: photometric fidelity, geometric consistency, and latent manifold smoothness.

### 4.5.1 Photometric Fidelity Constraints

The foundation of our reconstruction is ensuring that the rendered frames match the input observations. We formulate the appearance loss $\mathcal{L}_{\text{pho}}$ as a combination of pixel-wise accuracy and structural similarity to capture both high-frequency textures and local luminance structures:

$$\mathcal{L}_{\text{pho}} = (1-\lambda_{\text{ssim}})\|\mathcal{C}-\mathcal{C}_{gt}\|_1 + \lambda_{\text{ssim}}\big(1-\text{SSIM}(\mathcal{C}, \mathcal{C}_{gt})\big) \tag{19}$$

where $\mathcal{C}$ and $\mathcal{C}_{gt}$ represent the rendered and ground-truth image patches, respectively.

### 4.5.2 Monocular Geometric Distillation

A critical ambiguity in monocular dynamic scenes is the lack of absolute scale constraints. To prevent the "floater" artifacts where depth is sacrificed for texture fitting, we distill geometric priors from an off-the-shelf monocular depth estimator [55]. Rather than enforcing absolute depth values, which may be inconsistent across frames, we maximize the *Pearson correlation* $\rho$ between the rendered depth $D$ and the pseudo-ground truth $D_{\text{prior}}$. This ensures scale-and-shift invariant consistency:

$$\begin{aligned}\mathcal{L}_{\text{geo}} = \min\big(&1 - \rho(-D_{\text{prior}}, D),\\ &1 - \rho(1/(D_{\text{prior}} + 200), D)\big),\end{aligned} \tag{20}$$

where $\rho(\cdot,\cdot)$ denotes the Pearson correlation coefficient. This formulation is robust to global scale ambiguity and constrains the Chebyshev field to capture physically consistent global motion, while preventing the residual field from compensating for large-scale geometric errors.

### 4.5.3 Robust Latent Manifold Regularization

To stabilize the transformation field, we enforce local smoothness on the learned semantic embeddings $\mathbf{e}_s$. However, standard L2 regularization tends to over-smooth sharp semantic boundaries (e.g., the distinct transition between a moving hand and the background).

We enforce that the embeddings of spatially adjacent primitives remain coherent, while allowing for necessary discontinuities at object boundaries:

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{reg}} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} \sqrt{\omega_{ij} \|\mathbf{e}_i - \mathbf{e}_j\|_2 + \varepsilon} \quad (21)$$

where $\mathcal{N}(i)$ denotes the $k$-nearest neighbors of primitive $i$. The spatial weight $\omega_{ij} = \exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/\sigma^2)$ acts as a bilateral filter. The constant $\varepsilon$ ensures numerical stability during backpropagation. This robust formulation promotes *intra-object rigidity* (smoothing noise within an object) while preserving *inter-object separability* (maintaining sharp semantic edges).

# 5 Experimental

## 5.1 Implementation Details

Our framework is implemented in PyTorch. We employ the RAdam optimizer [30] for training, which provides robust convergence without the need for warm-up. The deformation field and 3D Gaussian parameters are optimized end-to-end. We train our model for $30,000$ iterations. The initial learning rates for position and rotation are set to $[1.6 \times 10^{-4}]$ and $[1 \times 10^{-3}]$, respectively.

## 5.2 Datasets and Metrics

**Datasets**. To evaluate the robustness and effectiveness of our proposed method, we conduct experiments on two challenging dynamic scene benchmarks:

- HyperNeRF Dataset [34]: This dataset focuses on topologically varying scenes captured with a single camera. It presents significant challenges due to complex motions and long-duration sequences.
- NeRF-DS Dataset [48]: This dataset features dynamic specular objects. It is particularly demanding as it requires the model to handle both geometric deformation and view-dependent appearance changes simultaneously.

For NeRF-DS, HyperNeRF, and other Nerfies-format datasets, we follow the official train/test splits specified in their dataset metadata. For multi-view video datasets such as Neural 3D Video, we adopt the commonly used leave-one-camera-out protocol, where one camera is held out for testing and the remaining cameras are used for training. This ensures that all methods are evaluated under standard train/test protocols consistent with prior work.

**Evaluation Metrics**. We quantitatively evaluate the reconstruction quality using three standard metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) for measuring pixel-level signal fidelity. To assess visual quality from a perceptual perspective, we utilize the Learned Perceptual Image Patch Similarity (LPIPS) [54]. LPIPS leverages deep feature extraction to evaluate image similarity; following standard protocols, we compute LPIPS using features extracted from the AlexNet [18] or VGG [39] architecture. We compare our method against state-of-the-art dynamic view synthesis approaches. For methods lacking public code or pretrained models, where comparable reproduction is not feasible, their corresponding results are marked with $-$ in our quantitative tables.

## 5.3 Quantitative Result on NeRF-DS Dataset

To comprehensively evaluate the reconstruction quality of our proposed method, we conduct a per-scene quantitative comparison on the challenging NeRF-DS dataset, which features complex specular reflections and dynamic topological changes. Table 1 reports the results in terms of PSNR, SSIM, and LPIPS metrics.

Overall Performance: As demonstrated in the `Mean` column of Table 1, our method achieves state-of-the-art (SOTA) performance across all three aggregated metrics, significantly outperforming existing baselines. Specifically, our method achieves an average PSNR of **24.58** dB and SSIM of **0.862**, surpassing the second-best method, D3DGS [52], by margins of 0.47 dB and 0.01, respectively. This indicates that our approach recovers scene geometry and photometric details with higher fidelity.

Performance in Perceptual Quality: The most prominent advantage of our framework lies in the Learned Perceptual Image Patch Similarity (LPIPS) metric, which aligns closely with human visual perception. Our method achieves a remarkable average LPIPS score of 0.132, which

| Method | As | | | Basin | | | Bell | | | Cup | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ |
| 3D-GS [17] | 22.69 | 0.802 | 0.299 | 18.42 | 0.717 | 0.315 | 21.01 | 0.789 | 0.250 | 21.71 | 0.830 | 0.255 |
| TiNeuVox [7] | 21.26 | 0.829 | 0.397 | 20.66 | 0.815 | 0.269 | 23.08 | 0.824 | 0.257 | 19.71 | 0.811 | 0.364 |
| HyperNeRF [34] | 25.58 | 0.895 | 0.178 | 20.41 | 0.820 | 0.191 | 23.06 | 0.810 | 0.205 | 24.59 | 0.877 | 0.165 |
| NeRF-DS [48] | 25.13 | 0.878 | 0.174 | 19.96 | 0.817 | 0.186 | 23.19 | 0.821 | 0.187 | 24.91 | 0.874 | 0.174 |
| D3DGS [52] | 26.31 | 0.884 | 0.178 | 19.67 | 0.793 | 0.190 | 25.74 | 0.850 | 0.154 | 24.86 | 0.891 | 0.153 |
| SCGS [14] | 24.94 | 0.859 | 0.221 | 18.45 | 0.729 | 0.281 | 20.04 | 0.704 | 0.334 | 20.09 | 0.775 | 0.315 |
| ADCGS [13] | 26.26 | 0.854 | 0.183 | 19.75 | 0.768 | 0.180 | 24.61 | 0.827 | 0.179 | 24.37 | 0.878 | 0.163 |
| 4DGS [44] | 25.58 | 0.861 | 0.165 | 19.62 | 0.783 | 0.156 | 25.55 | 0.861 | 0.132 | 24.40 | 0.874 | 0.147 |
| Ours | 26.76 | 0.884 | 0.139 | 19.98 | 0.806 | 0.147 | 26.14 | 0.889 | 0.091 | 25.03 | 0.898 | 0.106 |

| Method | Plate | | | Press | | | Sieve | | | Mean | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ | PSNR↑ | SSIM↑ | LPIPS-V↓ |
| 3D-GS [17] | 16.14 | 0.697 | 0.409 | 22.89 | 0.816 | 0.290 | 23.16 | 0.820 | 0.225 | 20.29 | 0.782 | 0.292 |
| TiNeuVox [7] | 20.58 | 0.803 | 0.332 | 24.47 | 0.861 | 0.300 | 21.49 | 0.827 | 0.318 | 21.61 | 0.823 | 0.278 |
| HyperNeRF [34] | 18.93 | 0.771 | 0.294 | 26.15 | 0.890 | 0.196 | 25.43 | 0.880 | 0.165 | 23.45 | 0.849 | 0.199 |
| NeRF-DS [48] | 20.54 | 0.804 | 0.200 | 25.72 | 0.862 | 0.205 | 25.78 | 0.890 | 0.147 | 23.60 | 0.849 | 0.182 |
| D3DGS [52] | 20.48 | 0.812 | 0.222 | 26.01 | 0.865 | 0.191 | 25.70 | 0.872 | 0.150 | 24.11 | 0.852 | 0.177 |
| SCGS [14] | 17.75 | 0.734 | 0.350 | 21.13 | 0.777 | 0.297 | 23.21 | 0.813 | 0.214 | 20.80 | 0.770 | 0.287 |
| ADCGS [13] | 20.56 | 0.784 | 0.244 | 25.86 | 0.827 | 0.208 | 25.59 | 0.857 | 0.164 | 23.86 | 0.828 | 0.189 |
| 4DGS [44] | 20.03 | 0.762 | 0.250 | 26.41 | 0.873 | 0.138 | 25.46 | 0.842 | 0.177 | 23.86 | 0.837 | 0.166 |
| Ours | 21.02 | 0.810 | 0.190 | 26.46 | 0.858 | 0.144 | 26.66 | 0.890 | 0.108 | 24.58 | 0.862 | 0.132 |

**Table 1**: Results on NeRF-DS [48]. Metrics: PSNR↑, SSIM↑, LPIPS↓. Best and second-best are in **red** and **orange**.

| Model | Average | | Broom | | 3D Printer | | Chicken | | Banana | |
|---|---|---|---|---|---|---|---|---|---|---|
| | PSNR↑ | SSIM↑ | PSNR↑ | SSIM↑ | PSNR↑ | SSIM↑ | PSNR↑ | SSIM↑ | PSNR↑ | SSIM↑ |
| Nerfies[1] [33] | 22.23 | - | 19.30 | - | 20.00 | - | 26.90 | - | 23.30 | - |
| HyperNeRF[1] [34] | 22.29 | 0.598 | 19.51 | 0.210 | 20.04 | 0.635 | 27.46 | 0.828 | 22.15 | 0.719 |
| TiNeuVox[1] [7] | 24.20 | 0.616 | 21.28 | 0.307 | 22.80 | 0.725 | 28.22 | 0.785 | 24.50 | 0.646 |
| D3DGS[1] [52] | 22.40 | 0.598 | 20.48 | 0.313 | 20.38 | 0.644 | 22.64 | 0.601 | 26.10 | 0.832 |
| E3DGS[1] [1] | 25.43 | 0.697 | 21.84 | 0.371 | 22.34 | 0.715 | 28.75 | 0.836 | 28.80 | 0.867 |
| ADCGS [13] | 25.31 | 0.696 | 21.83 | 0.361 | 22.97 | 0.738 | 29.78 | 0.871 | 26.64 | 0.816 |
| 4DGS [44] | 25.17 | 0.686 | 22.06 | 0.367 | 22.01 | 0.706 | 28.66 | 0.815 | 27.96 | 0.855 |
| Ours | 25.93 | 0.705 | 21.84 | 0.347 | 23.11 | 0.740 | 29.94 | 0.875 | 28.86 | 0.864 |

**Table 2**: Results on HyperNeRF [34]. Metrics: PSNR↑, SSIM↑. Best and second-best are in **red** and **orange**. Note: Results marked with [1] are taken from E3DGS [1].

is a **20.5**% reduction compared to the strongest baseline, 4DGS [44] (0.166), and a **25.4**% reduction compared to D3DGS [52]. This substantial gap highlights our model's exceptional capability in suppressing high-frequency artifacts and preserving sharp, realistic textures that are often smoothed out by other methods.

Robustness Across Diverse Scenes: Our method maintains strong and consistent performance across varying scene types. In scenes with complex occlusion and fine structures, such as `Plate` and `Sieve`, our method ranks first in all metrics. Notably, in the `Sieve` scene, we surpass 4DGS [44] by **1.2 dB** in PSNR and reduce LPIPS by nearly **39**% (0.108 vs. 0.177), proving our method's robustness in handling intricate geometries. In specular-dominant scenes like `Bell` and `Cup`, our method achieves the lowest LPIPS scores (0.091 and 0.106, respectively). This suggests that our approach effectively disentangles dynamic appearance changes (e.g., moving specular highlights) from geometry, minimizing the `foggy` or blurry artifacts common in NeRF-based baselines like HyperNeRF [34] or TiNeuVox [7]. In summary, while some baselines (e.g., D3DGS [52]) achieve competitive results in specific metrics like SSIM, our method provides the most balanced and visually pleasing reconstruction, establishing a new benchmark for dynamic scene synthesis on the NeRF-DS dataset.

## 5.4 Quantitative Result on HyperNeRF Dataset

As shown in Table 2, our method sets a new benchmark with an average PSNR of 25.93 dB and SSIM of 0.705, consistently outperforming recent dynamic Gaussian splatting methods such as E3DGS [1], ADCGS [13], and 4DGS [44].

While aggregate metrics provide a high-level overview, a granular analysis of the HyperNeRF dataset [34] reveals how different frameworks handle distinct categories of dynamic challenges, ranging from rigid motion to complex topological breaks.

**Handling Rapid Articulated Motion**: The `Chicken` scene represents a highly challenging scenario characterized by rapid, non-rigid articulated deformations. As shown in Table 2, conventional deformation-based methods often struggle here due to the difficulty of mapping large displacements continuously. Notably, our framework achieves a PSNR of 29.94$dB$, surpassing the recent E3DGS [1] and 4DGS [44] by margins of 1.19$dB$ and 1.28$dB$, respectively. This substantial performance gap ($> 1.0dB$) indicates that our Chebyshev transformation field, which explicitly modeling global trends, prevents the motion artifacts and geometric tearing that typically degrade the performance of continuous deformation fields in fast-moving articulated scenarios.

**Robustness to Topological Discontinuities**: The `3D Printer` scene introduces topological changes (material deposition) that violate the standard canonical-space assumptions of many NeRF-based methods. In this regime, our method achieves a PSNR of 23.11 dB and SSIM of 0.740 , significantly outperforming 4DGS [44] (22.01 dB /0.706). While 4DGS [44] relies on 4D primitives that may struggle to represent appearing or disappearing mass without artifacts, our `Divide-and-Conquer` strategy allows the RM field to implicitly compensate for these nonlinear topological breaks, maintaining geometric integrity where other methods fail.

**The Stability-Plasticity Trade-off**: We observe a nuanced trade-off in the `Broom` scene, which features relatively rigid motion. Here, 4DGS [44] achieves a slightly higher PSNR (22.06 dB) compared to ours (21.84 dB). However, this marginal advantage for 4DGS [44] in a rigid scene comes at the cost of severe degradation in non-rigid scenes like `Chicken` (where it trails ours by 1.28 dB). In contrast, our method demonstrates superior stability and generalization. It sacrifices a negligible amount of precision in simple rigid cases to achieve massive gains in complex dynamic scenarios. This confirms that our framework does not overfit to specific motion priors (e.g., rigidity) but instead learns a generalized representation capable of adapting to diverse dynamic characteristics.

Ultimately, our method is the only method that maintains high fidelity across the spectrum of dynamics—from the fine details of `Banana` (28.86 dB) to the articulated chaos of `Chicken`. This balanced performance is reflected in the highest average SSIM of 0.705, suggesting that our spectral decomposition approach provides more structurally consistent reconstruction for open-world dynamics.

## 5.5 Quantitative Results on N3DV Dataset

| Method | PSNR↑ | SSIM↑ | LPIPS−A ↓ |
|---|---|---|---|
| Mix Voxels[1] [40] | 30.30 | 0.918 | 0.127 |
| K-Planes[1] [8] | 30.86 | 0.939 | 0.096 |
| D3DGS[1] [52] | 28.27 | 0.917 | 0.156 |
| ST-GS[1] [26] | 28.94 | 0.920 | 0.078 |
| Swift4D[1] [45] | 30.05 | 0.931 | 0.055 |
| 4DGaussians[1] [6] | 30.30 | 0.933 | 0.069 |
| 4DGS[1] [44] | 30.19 | 0.917 | 0.061 |
| E-D3DGS[1] [1] | 30.79 | 0.937 | 0.051 |
| FRoG[1] [15] | 31.01 | 0.941 | 0.050 |
| 4K4D[2] [46] | 21.29 | 0.827 | – |
| ENeRF[2] [29] | 23.48 | 0.894 | – |
| 3DGS[2] [17] | 28.61 | 0.950 | – |
| Dynamic 3D Gaussians[2] [31] | 25.91 | 0.881 | – |
| LongVol-TemporalGH[2] [47] | 29.44 | 0.945 | – |
| DeGauss [42] | 31.31 | 0.942 | 0.094 |
| Ours | 31.58 | 0.943 | 0.063 |

**Table 3**: Quantitative comparison on the N3DV dataset [22]. Note: [1] denotes results reported by FRoG [15], and [2] denotes results reported by LongVol-TemporalGH [47].

We also evaluate our method on the Neural 3D Video (N3DV) dataset [22] to verify its generality on multi-view dynamic scenes. As shown in Table 3, our method achieves the best PSNR of 31.58, outperforming recent competitive methods such as FRoG [15] and DeGauss [42]. Compared with FRoG [15] , our method improves PSNR

from 31.01 to 31.58. Compared with DeGauss [42], our method further improves PSNR from 31.31 to 31.58 and reduces LPIPS from 0.094 to 0.063. These results demonstrate that our method is not limited to monocular videos, but also generalizes well to multi-view dynamic view synthesis.

It is worth noting that different methods show advantages on different metrics. For example, 3DGS reports the highest SSIM, and FRoG [15] achieves the best LPIPS. Nevertheless, our method achieves the best PSNR while maintaining competitive SSIM and LPIPS, suggesting a favorable balance between reconstruction fidelity and perceptual quality. Since part of the compared results are taken from previously reported numbers, we explicitly mark their sources in the table for transparency. Overall, the results on N3DV datasets further demonstrate the robustness and general applicability of our method across different dynamic scene settings.

## 5.6 Quantitative Result on iPhone Dataset

To further evaluate the effectiveness of our method under challenging monocular dynamic capture, we conduct additional experiments on the Dycheck iPhone dataset [9]. This dataset contains casual real-world videos with hand-held camera motion, non-rigid object deformation, and complex temporal variations. We compare our method with several representative dynamic Gaussian splatting methods, including SC-GS [14], D3DGS [52], 4DGS [44], and MoDec-GS [19]. As shown in Table 4, our method achieves the best average performance with 16.33 mPSNR, 0.576 mSSIM, and 0.356 mLPIPS. Compared with the second-best average mPSNR result, our method improves the reconstruction quality from 14.60 to 16.33, demonstrating a clear advantage on in-the-wild monocular dynamic scenes.

In the per-scene results, our method achieves the best mPSNR across all seven scenes, including Apple, Block, Paper-windmill, Space-out, Spin, Teddy, and Wheel. The improvement is particularly evident in scenes with large deformations or complex appearance changes. For example, our method achieves an mPSNR of 18.68 on Apple, 17.05 on Paper-windmill, and 16.47 on Spin. In addition, our method consistently achieves the best mSSIM across all scenes, indicating that the proposed representation better preserves structural details under dynamic motion. In terms of perceptual quality, our method also achieves the lowest average mLPIPS.

## 5.7 Representation Capacity Analysis

Table 5 reports a comprehensive comparison of model compactness and sparsity across multiple scenes, including AS, Basin, Bell, Cup, Plate, Press, and Sieve, with results averaged in the Mean column. We evaluate three complementary metrics: the number of Gaussian primitives (GP), the memory footprint of Gaussian parameters (GP (MB)), and the total model size (Total (MB)), where lower values indicate more compact representations.

Across all evaluated scenes, our method consistently produces significantly fewer Gaussian primitives than existing approaches, including 4DGS [44] and SCGS [14] . For instance, on the AS and Basin scenes, our model reduces the GP count by more than an order of magnitude compared to 4DGS [44] , while also maintaining a substantially lower memory footprint. This trend is consistently observed across both relatively simple scenes (e.g., Cup, Press) and more structurally complex scenes (e.g., Bell, Plate), indicating that the proposed representation does not rely on scene-specific heuristics to achieve sparsity.

In terms of memory efficiency, our method achieves the lowest GP (MB) and Total (MB) across all scenes, with particularly pronounced improvements in scenes where baseline methods exhibit dense Gaussian distributions. On average, the total memory consumption is reduced to approximately one-eighth of 4DGS [44] and less than one-third of SCGS [14] , demonstrating that the proposed approach scales favorably with scene complexity. Importantly, this reduction is not achieved by aggressively pruning Gaussian parameters alone; rather, it reflects a more compact and structured parameterization that inherently discourages redundant primitives.

Overall, these results demonstrate that the proposed method achieves substantially improved sparsity and compactness without sacrificing representational capacity, providing strong evidence that the learned Gaussian distributions are both structurally economical and memory-efficient.

| Method | Apple | | | Block | | | Paper-windmill | | | Space-out | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** |
| SC-GS[1] [14] | 14.96 | 0.692 | 0.508 | 13.98 | 0.548 | 0.483 | 14.87 | 0.221 | 0.432 | 14.79 | 0.511 | 0.440 |
| D3DGS[1] [52] | 15.61 | 0.696 | 0.367 | 14.87 | 0.559 | 0.390 | 14.89 | 0.213 | 0.341 | 14.59 | 0.510 | 0.450 |
| 4DGS[1] [44] | 15.41 | 0.691 | 0.524 | 13.89 | 0.550 | 0.539 | 14.44 | 0.201 | 0.445 | 14.29 | 0.515 | 0.473 |
| MoDec-GS[1] [19] | 16.48 | 0.699 | 0.402 | 15.57 | 0.590 | 0.478 | 14.92 | 0.220 | 0.377 | 14.65 | 0.522 | 0.467 |
| **Ours** | 18.68 | 0.774 | 0.370 | 16.96 | 0.642 | 0.406 | 17.05 | 0.329 | 0.242 | 16.74 | 0.623 | 0.341 |

| Method | Spin | | | Teddy | | | Wheel | | | Average | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** | **mPSNR↑** | **mSSIM↑** | **mLPIPS↓** |
| SC-GS[1] [14] | 14.32 | 0.407 | 0.445 | 12.51 | 0.516 | 0.562 | 11.90 | 0.354 | 0.484 | 13.90 | 0.464 | 0.479 |
| D3DGS[1] [52] | 13.10 | 0.392 | 0.490 | 11.20 | 0.508 | 0.573 | 11.79 | 0.345 | 0.394 | 13.72 | 0.461 | 0.430 |
| 4DGS[1] [44] | 14.89 | 0.413 | 0.441 | 12.31 | 0.509 | 0.605 | 10.83 | 0.339 | 0.538 | 13.72 | 0.460 | 0.509 |
| MoDec-GS[1] [19] | 15.53 | 0.433 | 0.366 | 12.56 | 0.521 | 0.598 | 12.44 | 0.374 | 0.413 | 14.60 | 0.480 | 0.443 |
| **Ours** | 16.47 | 0.557 | 0.339 | 13.59 | 0.556 | 0.443 | 14.79 | 0.553 | 0.350 | 16.33 | 0.576 | 0.356 |

**Table 4**: **Quantitative results comparison on the iPhone dataset [9].** Each scene reports mPSNR(dB)↑, mSSIM↑, and mLPIPS↓. Note: [1] denotes results reported by MoDec-GS [19].

| Method | AS | | | Basin | | | Bell | | | Cup | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ |
| 4DGS [44] | 132354 | 31.30 | 169.68 | 132034 | 31.23 | 169.60 | 215542 | 50.98 | 190.39 | 165732 | 39.20 | 177.99 |
| SCGS [14] | 274829 | 74.44 | 77.33 | 255066 | 69.08 | 72.39 | 287786 | 77.95 | 80.60 | 434939 | 117.80 | 120.64 |
| Ours | 44342 | 15.90 | 20.97 | 33521 | 12.02 | 17.78 | 54456 | 19.53 | 25.87 | 31296 | 11.22 | 16.93 |

| Method | Plate | | | Press | | | Sieve | | | Mean | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ | GP↓ | GP(MB)↓ | Total(MB)↓ |
| 4DGS [44] | 266204 | 62.96 | 202.99 | 91280 | 21.59 | 159.46 | 171203 | 40.49 | 179.34 | 167764 | 39.68 | 178.49 |
| SCGS [14] | 256661 | 69.52 | 72.13 | 128881 | 34.91 | 37.53 | 199986 | 54.17 | 56.81 | 262593 | 71.12 | 73.92 |
| Ours | 27030 | 9.69 | 20.83 | 48920 | 17.54 | 23.88 | 38780 | 13.91 | 20.91 | 39764 | 14.26 | 21.02 |

**Table 5**: Representation Capacity Analysis.

## 5.8 Analysis of Gaussian Point Evolution

Figure 3 presents a detailed analysis of the evolution of Gaussian point counts across training iterations for different object categories, together with their peak values. This experiment aims to investigate how scene structure and object geometry influence the optimization dynamics of Gaussian-based representations.

**Training Dynamics Across Categories.** From the left panel, several consistent trends can be observed. For most categories, the number of Gaussian points initially increases during early training stages, followed by a gradual decline as optimization progresses. This behavior reflects a typical over-parameterization followed by consolidation process: the model first introduces a larger number of Gaussians to capture coarse geometry and appearance variations, and subsequently prunes or merges redundant points as the representation becomes more stable.

Notably, the Bell category exhibits the most pronounced growth, reaching its peak around 7k training steps before a sharp reduction. This suggests that objects with strong curvature, thin structures, and high-frequency geometric details require a substantially larger number of Gaussian primitives during early optimization to approximate complex surface variations. In contrast, categories such as Plate and As show more moderate growth and smoother decay, indicating that relatively planar or geometrically simple objects can be represented with fewer adaptive components.

**Category-Specific Stability and Convergence.** Another important observation is the variation in convergence stability across categories. Objects like Bell and Sieve maintain relatively higher Gaussian counts even in later training stages, implying that these categories contain persistent fine-scale structures or repetitive geometric patterns that are not easily compressed without sacrificing fidelity. On the other hand, Basin and Cup show a more aggressive reduction after the peak, suggesting stronger redundancy among Gaussian primitives once the global geometry has been established. Interestingly, the mean trend closely follows the behavior of mid-complexity categories, confirming that the overall optimization

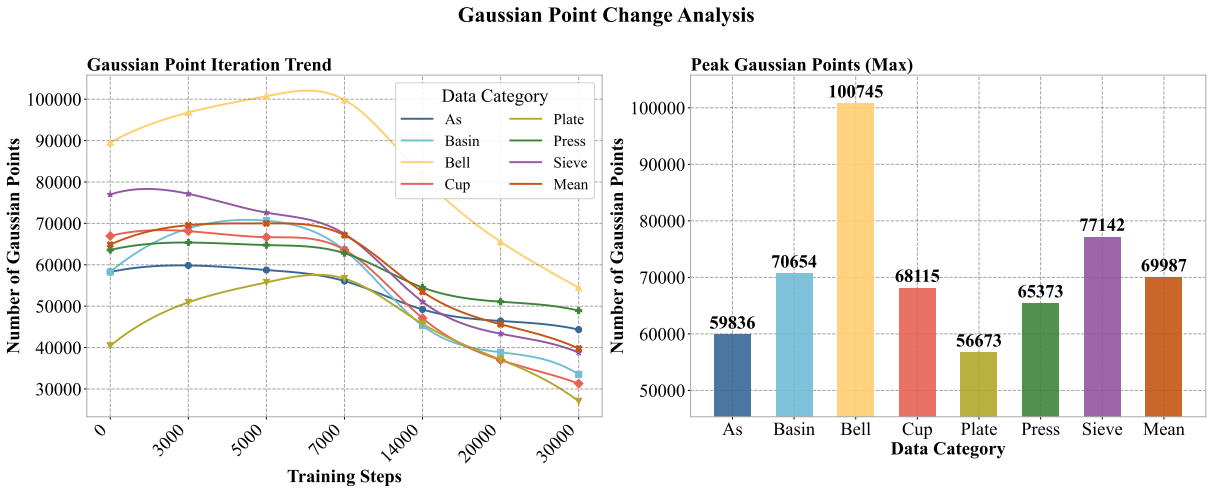



**Fig. 3**: Gaussian Point Change Analysis across Training. Evolution of the number of Gaussian points during training for different object categories (left), illustrating an initial expansion followed by category-dependent consolidation. The right panel reports the peak Gaussian count achieved for each category, revealing substantial variation in representation capacity requirements across object geometries. Objects with higher geometric complexity (e.g., Bell and Sieve) require significantly more Gaussian primitives, underscoring the necessity of deformation-field–based strategies rather than a uniform parameterization.

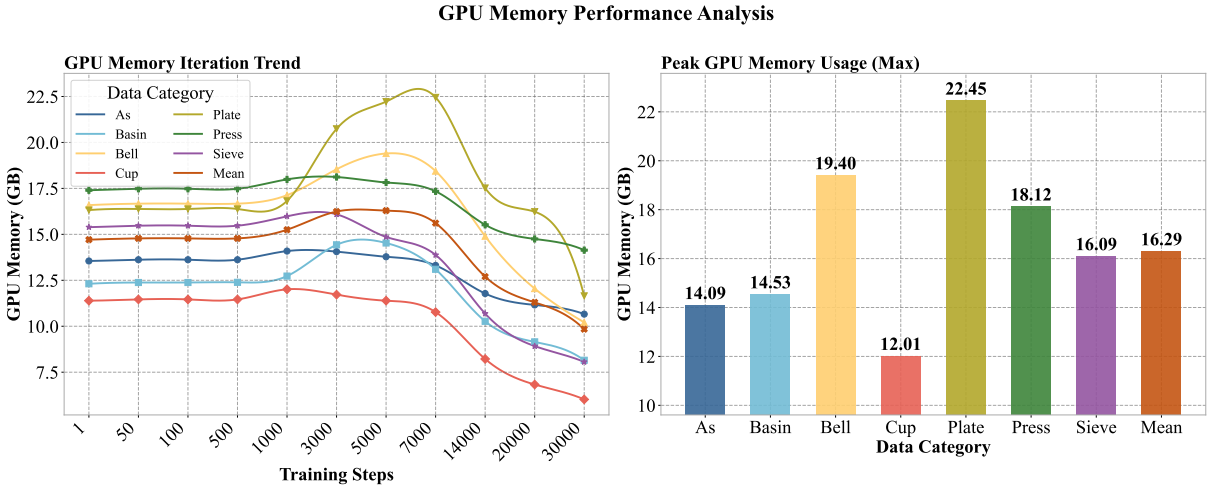



**Fig. 4**: GPU memory usage of the proposed method across scenes. Left: GPU memory consumption over training iterations. Right: Peak GPU memory usage during training.

strategy does not collapse to extreme overfitting or excessive pruning for any single object type.

**Peak Gaussian Count Analysis.** The right panel summarizes the maximum number of Gaussian points reached during training for each category. The peak values further support the above observations: Bell reaches the highest maximum (100,745 points), while Plate exhibits the lowest peak (56,673 points). This substantial gap quantitatively demonstrates how geometric complexity directly impacts the expressive capacity required by Gaussian-based representations. More importantly, the ordering of peak values correlates well with intuitive shape complexity rather than object scale alone, indicating that the model adaptively

allocates representation capacity based on structural difficulty rather than relying on a fixed budget.

**Implications for Adaptive Representation Design.** These results highlight a critical limitation of static or uniformly parameterized Gaussian representations. Since different object categories exhibit distinct capacity requirements and convergence behaviors, a one-size-fits-all parameterization is inherently suboptimal. The observed dynamics strongly motivate adaptive Gaussian allocation mechanisms, such as structure-aware initialization, category-conditioned pruning, or temporally adaptive point evolution strategies.

In summary, this experiment demonstrates that Gaussian point optimization is highly object-dependent, and effective dynamic scene modeling requires representations that can flexibly expand and contract in response to geometric complexity and training stage. This insight provides strong empirical justification for introducing adaptive or learnable point evolution strategies in Gaussian-based scene representations.

## 5.9 GPU Memory Usage

Figure 4 presents the GPU memory consumption behavior of the proposed method across different scenes during training, reporting both the iteration-wise memory trend and the peak GPU memory usage. This analysis aims to evaluate the stability and scalability of the model from a resource-efficiency perspective.

From the iteration-wise curves, GPU memory usage initially increases as the model progressively activates and refines Gaussian primitives during early and mid-stage training. This growth phase is consistently observed across all scenes, reflecting the gradual enrichment of the underlying representation. Notably, after reaching a scene-dependent peak, memory consumption monotonically decreases in later training stages. This downward trend indicates that redundant or low-contribution primitives are effectively consolidated or removed, resulting in a progressively more compact representation without abrupt fluctuations. The smoothness of the curves suggests that the memory reduction process is stable and does not introduce optimization instabilities.

The peak memory statistics further highlight the robustness of the proposed approach across scenes of varying complexity. While more challenging scenes such as Plate and Bell exhibit higher peak memory usage due to richer geometric and appearance variations, the overall memory footprint remains bounded within a moderate range. Importantly, simpler scenes (e.g., Cup and As) consistently require substantially less memory, demonstrating that the model adapts its resource allocation according to scene complexity rather than relying on a fixed, over-parameterized configuration.

Moreover, the relatively narrow gap between scene-specific peaks and the average memory usage indicates that extreme memory spikes are effectively avoided. This behavior contrasts with methods that rely on dense or static parameterization, where memory usage typically remains high throughout training. Instead, the proposed approach dynamically balances representation fidelity and memory efficiency, making it particularly suitable for long training schedules and large-scale dynamic scene modeling.

Overall, these results demonstrate that the proposed method achieves stable, adaptive, and memory-efficient training dynamics, aligning well with the practical deployment considerations emphasized in IJCV, especially for resource-constrained or large-scale applications.

## 5.10 Detail Reconstruction Analysis

Figure 5 presents a qualitative evaluation of our method against 4DGS [44] and ADCGS [13] on the NeRF-DS dataset, focusing on regions that exhibit diverse and challenging motion and appearance phenomena. The selected examples highlight several common sources of reconstruction errors in dynamic monocular scenes, including fine-scale texture loss, reflective material distortions, low-contrast illumination, and extreme fast motion.

Fine-scale texture and edge fidelity. Baseline methods struggle to maintain sharp contours and coherent textures in regions with intricate structures. For instance, skin edges and thin plant boundaries often appear blurred or ghosted in 4DGS [44] and ADCGS [13], accompanied by color bleeding in semi-transparent areas. By contrast, our approach consistently preserves high-frequency details and clear object boundaries,

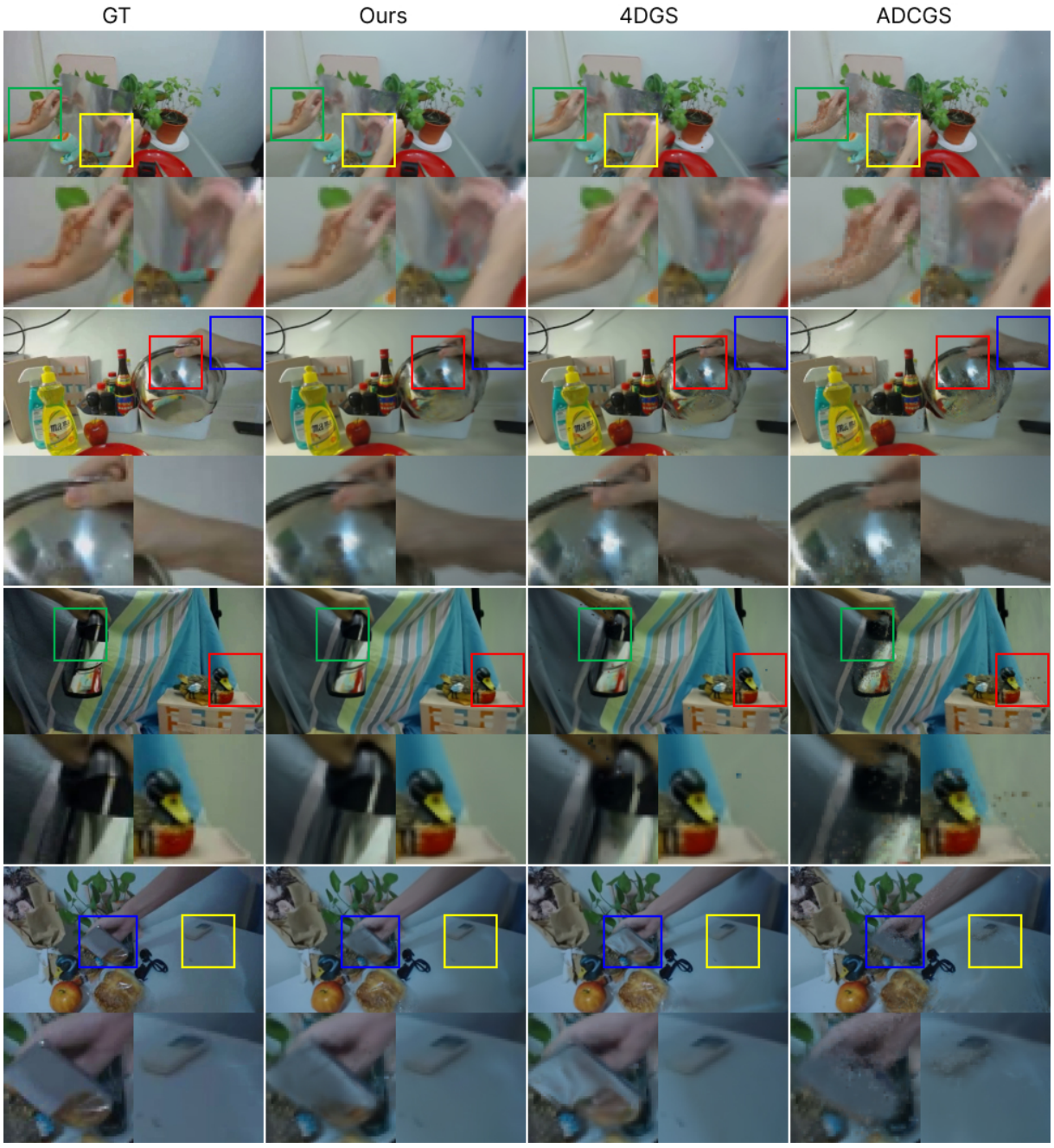



**Fig. 5**: Qualitative comparison with prior dynamic-scene methods demonstrates improved structural clarity and temporal consistency achieved by our approach.

demonstrating enhanced capability for reconstructing fine-scale textures under both moderate and rapid motion.

Reflective surfaces and specular highlights. Metallic or glossy materials pose significant challenges due to rapid appearance changes and highlight variations. In these regions, 4DGS [44] and ADCGS [13] frequently produce spot artifacts,

smearing, or noise amplification, which compromise micro-structural details. Our method, however, reliably captures specular reflections and subtle texture variations, maintaining geometric and photometric consistency even when motion is fast or surfaces are highly reflective.

Low-contrast and challenging illumination conditions. Scenes with subtle shading gradients or low-contrast lighting often lead to texture collapse or noise amplification in baseline reconstructions. ADCGS [13] is particularly prone to geometric distortions and excessive noise in these scenarios, while 4DGS [44] shows moderate artifacts. Our approach preserves smooth shading transitions without over-smoothing, ensuring realistic illumination and globally consistent appearance across both foreground and background regions.

Rapid motion and complex shadow–light interactions. High-speed movement and intricate lighting often induce motion blur, geometric inconsistencies, and texture tearing in baselines. Notably, objects such as handheld controllers and dynamic shadows are frequently misrepresented. Our method substantially mitigates these artifacts, producing stable geometry, accurate object contours, and coherent temporal transitions, even under extreme motion.

In summary, by organizing the evaluation around key visual challenges, the qualitative results demonstrate that our method consistently outperforms 4DGS [44] and ADCGS [13]. It excels in preserving fine textures, handling reflective surfaces, maintaining shading fidelity under low contrast, and ensuring temporal coherence under rapid motion. These observations complement the quantitative findings and underscore the robustness of our approach in capturing high-frequency details and complex dynamics in monocular dynamic scenes.

## 5.11 Ablation Study

***Component contribution analysis.***

We investigate how different design choices influence the reconstruction behavior of our framework through a comprehensive ablation study on the NeRF-DS dataset (in Table 6). Instead of treating individual components in isolation, we analyze how their removal alters the overall modeling capacity of the system.

| Method | PSNR↑ | SSIM↑ | LPIPS-V↓ |
|---|---|---|---|
| w/o Position | 22.46 | 0.804 | 0.213 |
| w/o Scales | 24.24 | 0.856 | 0.134 |
| w/o Rotations | 24.33 | 0.858 | 0.135 |
| w/o Depth Loss | 24.25 | 0.856 | 0.136 |
| w/ RM Field | 23.85 | 0.846 | 0.138 |
| w/ Chebyshev Field | 24.15 | 0.848 | 0.156 |
| Ours Full | 24.58 | 0.862 | 0.132 |

**Table 6**: We evaluate the contribution of individual components through an ablation study on the NeRF-DS [48] dataset.

A clear trend emerges when deformation modeling is progressively weakened. Suppressing positional deformation (*w/o Position*) causes the most severe degradation, with PSNR decreasing from 24.58 to 22.46 and SSIM from 0.862 to 0.804. This sharp decline indicates that without explicit spatial displacement, the model fails to establish reliable correspondences in dynamic scenes. Further degradation is observed when scale or rotation deformation is disabled, reflecting the model's reduced ability to accommodate non-rigid shape variations and orientation changes. These results collectively suggest that accurate dynamic reconstruction requires joint modeling of translation, scale, and rotation, rather than relying on a single geometric factor. We also examine the effect of replacing the proposed transformation field with alternative designs. Substituting our deformation field with RM or Chebyshev transformation fields consistently leads to inferior performance across all evaluation metrics. This performance gap highlights the inherent limitations of relying on a monolithic deformation field. A single transformation mechanism often struggles to simultaneously capture both global motion trends and high-frequency local variations.

In addition to deformation modeling, auxiliary supervision further influences reconstruction quality. Removing depth supervision (*w/o Depth Loss*) results in a mild but consistent performance drop, suggesting that depth information acts as a structural regularizer that stabilizes training and improves geometric consistency, even though it is not the primary driver of performance gains.

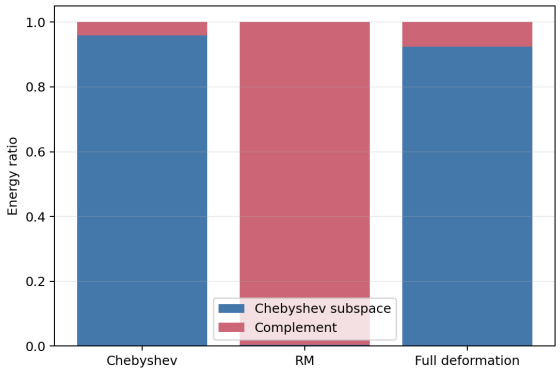



**Fig. 6**: **Chebyshev projection leakage analysis.** We project the Chebyshev branch, the RM branch, and the full deformation onto the predefined Chebyshev temporal subspace. Blue denotes the fraction of deformation energy explained by the Chebyshev subspace, while red denotes the fraction of energy lying in its complement. The RM branch is dominated by the complement component, indicating that it mainly captures residual deformation outside the low-order Chebyshev temporal subspace.

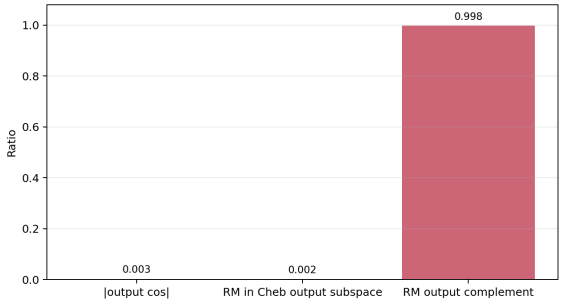



**Fig. 7**: **Output-level orthogonality diagnostics.** We measure the direct output overlap between the Chebyshev field and the RM field, and project the centered RM output onto the learned Chebyshev-output temporal subspace. The near-zero output overlap and low RM-to-Chebyshev-output leakage indicate that the RM branch has limited duplication of the Chebyshev output, while most RM energy remains in the complementary component.

***Chebyshev–RM complementarity analysis.*** To verify whether the RM branch captures complementary residual deformation rather than duplicating the Chebyshev branch, we conduct two diagnostics: Chebyshev projection leakage analysis and output-level orthogonality diagnostics. The former evaluates whether the learned deformation signals lie in the predefined Chebyshev temporal subspace or its complement, while the latter examines whether the RM output overlaps with the actual deformation pattern learned by the Chebyshev branch.

Fig. 6 shows the basis-level projection analysis. The horizontal axis denotes the three deformation signals, including the Chebyshev field, the RM field, and the full deformation, while the vertical axis denotes the normalized energy ratio in the Chebyshev temporal subspace and its complement. The Chebyshev branch is dominated by the Chebyshev-subspace component, confirming that it mainly models low-order temporal motion. In contrast, the RM branch has most of its energy in the Chebyshev-complement component, indicating that it mainly captures residual deformation outside the predefined Chebyshev temporal subspace. The full deformation contains both components, showing that the final motion combines the low-order Chebyshev trend and complementary RM residual motion.

Fig. 7 further evaluates the branch separation at the output level. Unlike the predefined Chebyshev basis used in Fig. 6, this diagnostic constructs a temporal subspace from the actual learned Chebyshev output and measures how much centered RM output energy is projected onto it. The near-zero output overlap and low RM-to-Chebyshev-output leakage indicate that the RM branch does not simply duplicate the learned Chebyshev deformation pattern. Instead, most RM energy remains in the complementary component, supporting the intended low-order-plus-residual motion decomposition.

## 5.12 Ablation Study: Region-wise Qualitative Analysis

As illustrated in Figure 8, comparing our full framework with variants that remove individual components reveals clear and systematic performance degradation patterns. These failure modes collectively validate the necessity of our overall design.

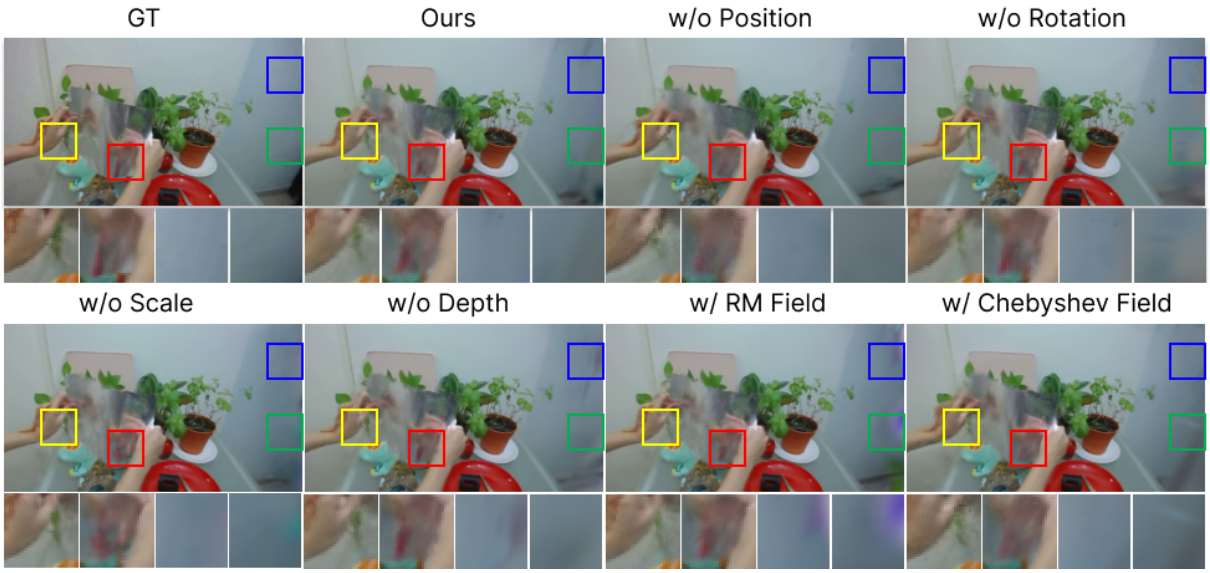



**Fig. 8**: Failure mode analysis of our method under component ablations. Visual comparison between the full framework and variants with individual components removed.

**Motion blur and fading artifacts**. Accurate recovery of dynamic trajectories relies on the synergy between explicit spectral decomposition and precise spatial alignment.

Effect of single-field designs: When the model relies solely on the RM field, it lacks the frequency-aware constraints provided by the Chebyshev bases. This deficiency manifests as blurred background reconstruction and noticeable color fading, such as the disappearance or desaturation of yellow box. These artifacts are characteristic of missing high-frequency details, confirming that purely implicit MLP-based modeling struggles to capture high-frequency information effectively. Effect of removing positional modeling: Similarly, removing the position attribute prevents the model from correctly tracking object centroids, leading to comparable blurring and fading artifacts.

**Geometric distortion and jitter**. Without accurate anisotropic modeling, structural consistency is severely compromised.

Effect of geometric priors: We observe that removing rotation or scale attributes leads to varying degrees of geometric failure, such as deformation of the hand region (red box). Moreover, without these priors, Gaussian primitives cannot be properly aligned with surface orientations, resulting in blurred planar surfaces (blue box) and pronounced background edge jitter (green box) caused by global misalignment.

**High-frequency noise and blending errors**. Suppressing residual artifacts and correctly handling occlusions critically depend on implicit reasoning and depth supervision.

Effect of single-field designs: When only the Chebyshev field is used, the model captures global trends but fails to resolve nonlinear dynamics, leading to noise-like artifacts on planar surfaces (blue box and green box). Conversely, when only the RM field is employed, the model can represent nonlinear motion but lacks the ability to capture high-frequency details, resulting in severe color blending errors in cluttered background areas. Effect of removing depth supervision: Similarly, eliminating depth supervision causes depth ambiguity and blur, further exacerbating color blending errors.

Only the full model, which integrates the spectral precision of Chebyshev, the local adaptability of RM, and robust geometric attributes, can effectively mitigate these artifacts while maintaining temporal stability and texture fidelity across all regions.

# 6 Discussion

A central contribution of this work lies not in introducing a more expressive deformation model, but in revisiting how dynamic motion should be represented. Our experiments suggest that many failure modes observed in monocular dynamic reconstruction, such as temporal jitter, boundary instability, and geometry collapse, are rooted in a monolithic modeling assumption that forces all motion into a single function space. By explicitly decomposing motion into a globally stable spectral component and a localized residual component, our method provides a principled way to balance temporal coherence and representational flexibility.

**Spectral Modeling.** The Chebyshev field highlights the advantage of explicit spectral modeling for global motion trends. Unlike implicit deformation networks, which are susceptible to spectral bias and optimization conflicts, Chebyshev-based representations offer strong inductive bias for bounded, non-periodic motion. Importantly, our formulation does not aim to replace implicit models entirely, but to restrict their role to residual dynamics where flexibility is truly required. This perspective suggests that explicit and implicit representations are complementary rather than competing choices.

**Residual Motion.** A key insight of this work is that residual motion components should not be treated as optimization artifacts or noise. Instead, they exhibit structured spatial and temporal patterns tied to object identity, articulation, and local interactions. The proposed Residual Motion Field explicitly models this structure while remaining decoupled from global motion trends. This separation not only improves stability but also enhances interpretability, as different motion regimes are handled by dedicated components.

**Limitations.** Despite its advantages, the proposed framework has several limitations. First, the decomposition assumes that global motion can be well approximated by low-frequency spectral components, which may be insufficient for extremely abrupt or discontinuous global changes. Second, the residual motion field introduces additional parameters and optimization complexity, which increases memory consumption compared to purely static Gaussian splatting methods. Finally, while the framework is effective for monocular inputs, incorporating multi-view or multi-modal cues could further strengthen geometric consistency.

**Future Directions.** The proposed motion decomposition framework opens several avenues for future research. One promising direction is to explore adaptive or data-driven spectral bases that dynamically adjust to scene motion statistics. Another direction is to extend the residual motion field with physical or kinematic constraints to further improve robustness. Finally, integrating the framework with multi-view, depth, or inertial sensing could enable more accurate and scalable dynamic scene reconstruction.

# Conflict of Interest

The authors declare that they have no conflict of interest.

# Funding

This work was supported by the National Natural Science Foundation of China (No. 62072388), the Fujian Provincial Science and Technology Major Project (No. 2024HZ022003), the Jiangxi Provincial Natural Science Foundation Key Project (No. 20244BAB28039), the Xiamen Public Technology Service Platform (No. 3502Z20231043), and the Fujian Sunshine Charity Public Welfare Foundation.

# Data Availability

The datasets analysed during the current study are publicly available benchmarks. The HyperNeRF dataset is available in Ref. [34], N3DV dataset is available in Ref. [22], Dycheck iPhone dataset is available in Ref. [9], and the NeRF-DS dataset is available in Ref. [48].

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